Inorganic Cumulative Exam

November 11, 2014

The following questions require no prior knowledge of the chemistry of ICl_3 but instead rely on a simple understanding of spectroscopy and simple chemistry for the answers.

1. Draw the Lewis Dot structure of ICl_3, give a drawing and description of the expected geometry of the molecule and assign it to a point group?

2. ICl_3 is produced by the simple combination of I_2 and Cl_2 under conditions that minimize the production of other species. IR is a good tool for monitoring this reaction. Consider the IR spectra of the product and reactants. How many bands would you expect to see for each? You do not have to go through the tedious work of creating an irreducible representation to answer the question. The symmetry will tell you all that you need.

3. Mass spectroscopy is also an excellent tool for the investigation of the reaction. What would you expect to see besides the parent ion that would be diagnostic for the number of chlorine atoms in the molecule?

4. There is considerable evidence that ICl_3 is actually dimeric, I_2Cl_6. Draw the Lewis Dot structure of I_2Cl_6, give a drawing and description of the expected geometry of the molecule and assign it to a point group?

5. Molten I_2Cl_6 (or ICl_3) has an appreciable conductance. Suggest an explanation for that observation.